

**Shyamoli Engineering College**  
**Dept. of Computer Science and Engineering**  
**1<sup>st</sup> Year, 2<sup>nd</sup> Semester, 1<sup>st</sup> Assessment Examination**  
**Course Title: Fundamentals of Programming**  
**Course Code: CSE-1202**

Answer any four from the following Questions

**Full Mark-30**

**Time- 1 Hour**

1. What is a function in C? Write down the Difference between formal argument and actual argument with proper example? 5
2. Differentiate between a - - and - - a with proper example. 5  
Evaluate the following expressions:  
i)  $22 + 3 < 6 \ \&\& \ !5 \ || \ 22 == 7 \ \&\& \ 22 - 2 > +5$   
ii)  $a + 2 > b \ || \ !c \ \&\& \ a == d * a - 2 < = e$   
Where  $a=11$ ,  $b=6$ ,  $c=0$ ,  $d=7$  and  $e=5$
3. Write a program in C to check whether a number is Odd or Even using Function. 5
4. Explain different classification of user defined functions based on parameter passing and return type with examples 5
5. Write a c-program using function to check whether the given number is prime or not. 5

**Shyamoli Engineering College**  
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**1<sup>st</sup> Year, 2<sup>nd</sup> Semester, 1<sup>st</sup> Assessment Examination**  
**Course Title: Digital Logic Design**  
**Course Code: CSE-1202**

Answer any **four** from the following Questions

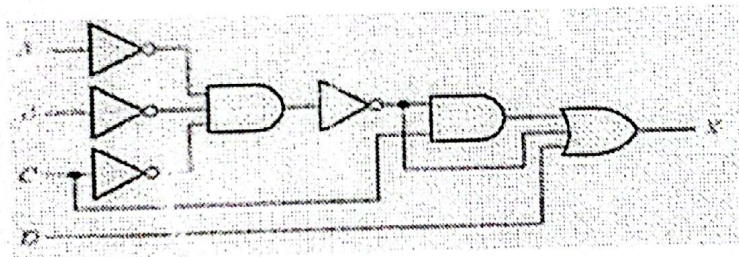
**Full Mark-30**

**Time- 1 Hour**

1. Using Boolean Algebra rules, do the following 5  
a) Simplify the following logic expression

$$f = AB + A\bar{C} + C + AD + A\bar{B}C + ABC$$

- b) Simplify the logic circuit in to a minimum form



2. Show the truth table for the following function 5

$$f(w, x, y, z) = wx + xz + \bar{y}$$

3. Use K-map to simplify the Boolean Expression: - 5

$$E = \bar{A}\bar{B}\bar{C}D + \bar{A}CD + \bar{A}\bar{C} + C$$

4. What are the various logic gates, give the representation along with the truth table. 5

5. Distinguish between canonical and standard forms by giving an example. Convert the following to the corresponding bases 5

i)  $(700)_{16} = ( )_8$

ii)  $(F7A9)_{16} = ( )_2$

**Shyamoli Engineering College**  
**1<sup>st</sup> assessment Examination**  
**Sub: Physics; PHY-1203**  
**Department: CSE**  
**1<sup>st</sup> YEAR 2<sup>nd</sup> SEMESTER**

**Full Mark-20**

**Time- 1hr**

- |                                                                                                                                                                                                                                      |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1. (a) Write down the difference between reversible and irreversible process.                                                                                                                                                        | 03 |
| (b) What do you mean by Carnot cycle? Derive the cycle with it's working principle.                                                                                                                                                  | 08 |
|                                                                                                                                                                                                                                      |    |
| 2. (a) State the 1 <sup>st</sup> law of thermodynamics.                                                                                                                                                                              | 02 |
| (b) Show that, In the irreversible cycle there is a net increase in entropy.                                                                                                                                                         | 04 |
| (c) A Carnot engine whose low temperature reservoir is at 7° C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased? | 03 |

# Shyamoli Textile Engineering College

Dept. of Computer Science & Engineering

Year-1, Sem-2, 1<sup>st</sup> Assessment Examination

Sub: Methods of Int. & Diff. Cal. & Series (MATH-1204)

Full Mark-25

Time- 1 Hour

(Answer all the following Questions)

1. a) Define order & degree of a ODE. Find the differential equation whose solution is  $v = \frac{A}{r} + B$  where A & B are arbitrary constant. (05)

- b) Show that the D.E of  $Ax^2 + By^2 = 1$  is  $x \left[ y \frac{d^2y}{dx^2} + \left( \frac{dy}{dx} \right)^2 \right] = y \frac{dy}{dx}$ . Hence find the order & degree. (05)

2. a) Solve  $\frac{dy}{dx} + \frac{y}{x} \ln y = \frac{y}{x^2} (\ln y)^2$ . (07)

- b) Find the general solution & singular solution of  $p = px + p - p^2$ . (08)

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24/11/21

20/11/21

**Amoli Textile Engineering College**  
**Dept. of Computer Science & Engineering**  
**1Y2S 1<sup>st</sup> Assessment Examination**  
**Course Title: Developing English Language Skills**  
**(ENG-1205)**

**Full Mark-25**

**Time- 1 Hour**

1. Last weekend, my friends and I went on \_\_\_\_ 5  
(1) trip to \_\_\_\_ (2) countryside. We started  
early in \_\_\_\_ (3) morning to enjoy \_\_\_\_ (4)  
fresh air and peaceful nature. On our way,  
we saw \_\_\_\_ (5) farmer working in his field  
with \_\_\_\_ (6) ox. After reaching \_\_\_\_ (7)  
riverside, we sat under \_\_\_\_ (8) big tree and  
had breakfast. It was \_\_\_\_ (9) wonderful  
experience to escape from \_\_\_\_ (10) noise of  
the city and spend some time close to nature.
2. Every morning, I wake up \_\_\_\_ (1) 6:30 a.m. 5  
and get ready \_\_\_\_ (2) school. I start my day  
\_\_\_\_ (3) brushing my teeth and taking a  
shower. After that, I sit \_\_\_\_ (4) the table to  
have breakfast. I usually eat bread and eggs  
\_\_\_\_ (5) a glass of milk. Then I put my books  
\_\_\_\_ (6) my bag and leave \_\_\_\_ (7) home. I  
walk \_\_\_\_ (8) the bus stop and wait \_\_\_\_ (9)  
the bus to arrive. My school starts \_\_\_\_ (10)  
8:00 a.m., and I always try to reach on time.
3. Every day, Rahim \_\_\_\_ (1) (get) up early in 5  
the morning. He \_\_\_\_ (2) (go) for a walk  
before the sun \_\_\_\_ (3) (rise). After returning  
home, he \_\_\_\_ (4) (take) a shower and \_\_\_\_  
(5) (have) his breakfast. Then he \_\_\_\_ (6)  
(leave) for his office which \_\_\_\_ (7) (be)  
about two kilometers away from his house.  
Yesterday, he \_\_\_\_ (8) (be) late because his  
alarm \_\_\_\_ (9) (not/ring). So, he \_\_\_\_ (10)  
(run) all the way to catch the bus.
4. Suppose you are a reporter of Daily Star. 10  
Write a report on the "Use of Artificial  
Intelligence in Education"